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United States Patent Application Publication

20250282651

Kind Code

A1

Publication Date

September 11, 2025

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SEPARATOR SYSTEM FOR USE IN AGRICULTURAL PROCESSING

Abstract

An agricultural processing system includes a process that uses water and produces wastewater having insoluble solids, a pump configured to receive and pressurize the wastewater from the process, and a plurality of centrifugal separators connected in parallel and configured to receive the wastewater from the pump. The separators are each configured to separate out the insoluble solids from the wastewater to produce a cleaned wastewater.

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Family ID: 1000008620906

Appl. No.: 19/207275

Filed: May 13, 2025

Related U.S. Application Data

parent US division 17454908 20211115 parent-grant-document US 12351489 child US 19207275

Publication Classification

Int. Cl.: C02F1/38 (20230101); A22C21/04 (20060101); C02F9/00 (20230101); C02F103/20 (20060101); C02F103/22 (20060101)

U.S. Cl.:

CPC C02F1/38 (20130101); A22C21/04 (20130101); C02F1/385 (20130101); C02F9/00 (20130101); C02F2103/20 (20130101); C02F2103/22 (20130101); C02F2209/005

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. Nonprovisional patent application Ser. No. 17/454,908 filed Nov. 15, 2021, the disclosure of which is hereby incorporated by reference in its entirety.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates to removal of insoluble materials from aqueous streams used in agricultural processing. More particularly, the application relates to removal of insoluble materials from aqueous streams used in agricultural processing using a centripetal force-based solid/liquid separator.

BACKGROUND

[0003] Poultry processing establishments transform a live animal to a whole bird or individual parts for consumption by the general population. The process of this transformation requires multiple unit operations that involve large volumes of water. During the various processing steps, the water streams become contaminated with solids such as feathers, ingesta, and fecal matter as well as insoluble liquids such as lipids.

[0004] In order to ensure a safer food supply for the general population, the United States Department of Agriculture (USDA) has established microbial performance standards for these establishments. Antimicrobial interventions are used at the various unit operations to aid in microbial reduction, allowing the establishments to achieve the performance standards. However, the presence of solid and liquid contaminants from the process can also interact with the antimicrobials, especially oxidizers such as peracetic acid (PAA), rendering the antimicrobials less effective. In addition, the solids, such as feathers and fecal matter, contain bacteria that can increase the microbial load that will need to be treated downstream. Solids, such as ingesta, and liquids, such as insoluble fat, can serve as food sources for microbes. These issues can increase the difficulty for establishments to achieve the USDA microbial performance standards.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Various embodiments of the present disclosure will be understood more fully from the detailed description given below and from the accompanying drawings. In the drawings, like reference numbers may indicate identical or functionally similar elements. Embodiments are described in detail hereinafter with reference to the accompanying figures, in which:

[0006] FIG. 1 is a schematic diagram of an agricultural processing system according to an embodiment of the present disclosure.

[0007] FIG. 2 is a schematic diagram of an agricultural processing system according to an embodiment of the present disclosure.

[0008] FIG. 3 is a graph depicting the results of Example 1.

[0009] FIG. 4 is a graph depicting the results of Example 2.

[0010] FIG. 5 is a graph depicting the results of Example 2.

[0011] FIG. 6 is a graph depicting the results of Example 3.

[0012] FIG. 7 is a graph depicting the results of Example 4.

[0013] FIG. **8** is a graph depicting the results of Example 4.

[0014] FIG. **9** is a graph depicting the results of Example 4.

[0015] FIG. **10** is a graph depicting the results of Example 4.

DETAILED DESCRIPTION

[0016] The following disclosure provides many different embodiments or examples. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed. While the present disclosure is described herein with reference to illustrative embodiments for particular applications, it should be understood that embodiments are not limited thereto. Other embodiments are possible, and modifications can be made to the embodiments within the spirit and scope of the teachings herein and additional fields in which the embodiments would be of significant utility.

[0017] Referring to FIG. **1**, a separator system **100** for use in agricultural processing is shown. The system includes an inlet supply **10**, which supplies a liquid to be treated. In one or more embodiments, the liquid is an aqueous suspension comprising one or more insoluble components. For example, the insoluble components may comprise feathers, fecal matter, ingesta, and/or fats. In one or more embodiments, the inlet supply **10** is in fluid communication with a process that uses water, such as a scalding tank, a food washing station such as an inside outside bird washer, a chiller or dip tank, or a combination thereof. The inlet supply **10** directs the liquid to a pump **12**, which pumps the fluid to a manifold **28** at a desired velocity and pressure.

[0018] In the system **100**, the liquid is directed from the manifold **28** through inlet lines **30a**, **30b**, **30c** to separators **18a**, **18b**, **18c** that are configured to clean the liquid by separating insolubles therefrom. The separators **18a**, **18b**, **18c** include underflow lines **34a**, **34b**, **34c** through which the insolubles exit the separators **18a**, **18b**, **18c** and overflow lines **32a**, **32b**, **32c** through which the cleaned liquid exits the separators **18a**, **18b**, **18c**. The insolubles may be directed to an underflow hopper **22** while the cleaned liquid is directed to an outlet **24** for use within the agricultural processing plant.

[0019] The pump **12** is configured to pump the liquid at a proper rate and/or pressure in order to obtain the centripetal force and centrifugal force needed to separate insolubles from the liquid in separators **18a**, **18b**, **18c**. In some embodiments, the pump **12** is a centrifugal pump capable of pumping at 10-250 gallons per minute (gpm), 50-200 gpm, 75-150 gpm, at least 10 gpm, at least 15 gpm, at least 25 gpm, at least 50 gpm, at least 75 gpm, or at least 100 gpm.

[0020] The connections between various components of the system **100** are not particularly limited and, in some embodiments, the connections may comprise tubing of various diameters and lengths. In some embodiments, a line **38** connecting the inlet supply **10** to the pump **12** comprises tubing that is about 4" in diameter proximate the inlet supply **10** and that reduces to about 2" in diameter proximate the pump **12**. In some embodiments, the pump **12** may output the liquid through line **42** having a diameter of about 2" that feeds into a plenum of the manifold **28** having a diameter of about 3". In some embodiments, inlet lines **30a**, **30b**, **30c** from the manifold **28** have a diameter of about ½". In some embodiments, overflow lines **32a**, **32b**, **32c** may comprise about ¾" tubing that manifolds into a line **40** having a diameter of about 3" that reduces to a diameter about 2" prior to reaching the outlet **24**.

[0021] The manifold **28** directs the liquid from the pump **12** to the plurality of separators **18a**, **18b**, **18c** via inlet lines **30a**, **30b**, **30c**. Although three separators **18a**, **18b**, **18c** are shown in FIG. **1**, the system **100** may include any number of separators. In some embodiments, the manifold **28** is excluded and the system **100** only includes a single separator. In other embodiments, the system includes two, four, five, six, seven, or eight or more separators. Each separator **18a**, **18b**, **18c** may be a centripetal solid/liquid separator that does not include any moving parts, which thereby

reduces maintenance costs. Suitable separators include those available from Compatible Components Corporation under the tradename VORSPIN. The separators **18a**, **18b**, **18c** may be suitably selected based on the operating needs of the system **100**. For example, larger separators **18a**, **18b**, **18c** may be able to process larger volumes of liquid, i.e., be operated at a greater operating pressure and process rate. However, in general, larger separators are not as efficient at removing small particulates. In some embodiments, the separators **18a**, **18b**, **18c** are capable of removing particulates having sizes of greater than about 7 microns, about 7 to about 18 microns, about 7 to about 10 microns, greater than about 10 microns, about 10 to about 12 microns, greater than about 12 microns, about 12 to about 17 microns, greater than about 15 microns, or about 15 to about 18 microns. In some embodiments, the separators **18a**, **18b**, **18c** are configured to operate at a pressure of 20-100 psig, 30-80 psig, 30-60 psig, psig, 30-40 psig, 40-45 psig, 40-50 psig, or 45-60 psig. In some embodiments, the separators **18a**, **18b**, **18c** are configured to operate at a process rate of at least 10 gpm, at least 20 gpm, at least 30 gpm, at least 40 gpm, at least 50 gpm, at least 60 gpm, 10-120 gpm, 20-100 gpm, 30-80 gpm, or 40-70 gpm. When connected in parallel as shown in FIG. 1, the separators **18a**, **18b**, **18c** typically have the same configuration since the liquid is supplied to the separators **18a**, **18b**, **18c** at the same rate from the pump **12**.

[0022] In one or more embodiments, the inlet supply **10** is a tank or other vessel having a set volume of liquid. In such embodiments, the system **100** may be configured based on a desired turnover rate of the volume of liquid. For example, for a 1500-gallon volume of liquid, three separators **18a**, **18b**, **18c** each operating at 50 gpm would process the entire volume (i.e., complete one turnover) in 10 minutes whereas operating only two of the same separators would take 20 minutes to complete one turnover. In one or more embodiments, the system **100** is configured to have a turnover rate of at least 0.1/hr, at least 0.25/hr, at least 0.5/hr, at least 1/hr, at least 1.5/hr, at least 2/hr, at least 3/hr, at least 4/hr, at least 5/hr, or at least 6/hr. Higher turnover rates generally lead to reduced microbial counts and may help reduce maintenance by maintaining a cleaner liquid.

[0023] After the liquid enters the separators **18a**, **18b**, **18c**, centrifugal separation occurs wherein an underflow of insoluble materials is expelled through underflow lines **34a**, **34b**, **34c**. The underflow lines **34a**, **34b**, **34c** may be manifolded and directed to an underflow hopper **22** for disposal or recycling. Cleaned fluid exits at the top of the separators **18a**, **18b**, **18c** through overflow lines **32a**, **32b**, **32c**. The cleaned fluid is then directed to an outlet **24**, which may be in fluid communication with the same process as that of the inlet supply **10** (i.e., may add back cleaned fluid to the process) or the cleaned fluid may be directed from the outlet **24** to another process within the agricultural processing plant as needed.

[0024] In one or more embodiments, the system **100** includes one or more inlet sensors for measuring conditions of the liquid being supplied into the system **100** and one or more outlet sensors for measuring conditions of the cleaned liquid being outputted by the system **100**. For example, the inlet sensors may include a flow and temperature sensor **14**, which may be positioned between the pump **12** and the separators **18a**, **18b**, **18c**. The inlet sensors may also include an inlet pressure sensor **16a** positioned between the pump **12** and the separators **18a**, **18b**, **18c**. The outlet sensors may include, for example, an outlet pressure sensor **16b** positioned after the separators **18a**, **18b**, **18c** to measure conditions of the cleaned liquid. The outlet sensors may also include a flow meter or temperature sensor (not shown). Each of the sensors may be in wired or wireless communication with a control panel (not shown), thereby providing real time information to a user. The real time information displayed on the control panel may include, for example, operating speeds (flow rates) and water savings based on the output rate of cleaned liquid. In some embodiments, the control panel may be partially or wholly automated, wherein measurements exceeding predetermined threshold values (too high or too low) cause one or more actions to be performed by the control panel. For example, an abnormal temperature reading may cause the control panel to produce an alarm or to shut down the system **100**. Excessive temperatures of liquids used for poultry processing may begin to melt fats or cook the poultry, which is undesirable.

Further, high temperatures might overheat the pump **12**. Conversely, in scalding operations, liquids that are too cool will cause feathers to not pick. In some embodiments, the inlet supply **10** is in communication with a chiller tank and the threshold maximum temperature is 50° F., 45° F., 40° F., or 35° F. In some embodiments, the inlet supply **10** is in communication with a scalding tank and the threshold maximum temperature is 150° F., 160° F., or 170° F. and the threshold minimum temperature is 110° F., 120° F., or 130° F.

[0025] As another example, an abnormal output pressure reading may cause the control panel to change the rate at which the pump **12** operates and/or produce an alarm or shut down the system **100**. If the measured pressure is too high or too low at the inlet or outlet of the system **100**, it could be indicative of a blockage within the system **100**. In some embodiments, a threshold minimum pressure measured at outlet pressure sensor **16b** is 5 psig, 4 psig, 3 psig, or 2 psig. In some embodiments, a threshold maximum pressure measured at outlet pressure sensor **16b** is 10 psig, 8 psig, or 6 psig.

[0026] In some embodiments, the control panel comprises a variable frequency drive (VFD) configured to control the speed of the pump **12**. The VFD may be manually or automatically operated. In an embodiment, a user would review data at the control panel to ensure that measurements from pressure sensors **16a** and **16b** are within predetermined ranges based on the specifications of the separators **18a**, **18b**, **18c**, verify that a flow rate measured by sensor **14** meets process demands, and then adjust the VFD as needed. In some embodiments, the VFD frequency is not to exceed 60 Hz.

[0027] In some embodiments, the system **100** includes an inlet sample port **20a** for sampling the liquid being provided into the system **100** and/or an outlet sample port **20b** for sampling the cleaned liquid being outputted by the system **100**. The sample ports **20a** and **20b** may be used for periodic quality control, wherein the sampling may be manual or automated. Sampling data may include, for example, microbial content, pH, solids content, and/or turbidity of the sampled liquid. In some embodiments, the sampling data may be used to establish Statistical Process Control (SPC) parameters, which may then be used to monitor the system **100**. In one or more embodiments, sampling is automated and sampling data is communicated to the control panel where it may be viewed by a user and/or may cause an automated response such as an alarm, for instance, when data values fall outside of established SPC parameters.

[0028] In one or more embodiments, the system **100** includes a supply **26** configured to inject an additive to the cleaned liquid. In some embodiments, the additive is an antimicrobial, a pH adjuster, or fresh water. The supply **26** may be operated as needed in view of the measured conditions of the system **100**.

[0029] Turning to FIG. 2, in an alternative system **200**, the separators **18d**, **18e** may be configured in series. Namely, in the system **200**, liquid from the inlet supply **10** is directed to first pump **12a**, which may be as described above. The liquid is transported from the first pump **12a** via line **30d** to the first separator **18d**. The liquid is cleaned in the first separator **18d** such that a cleaned overflow is directed from overflow line **32d** into a tank **36**. The liquid in tank **36** is then pumped by a second pump **12b**, which may be the same type as the first pump **12a**, via line **30e** to the second separator **18e**. Each of the separators **18d**, **18e** produce an underflow through underflow lines **34d**, **34e** which may be directed to an underflow hopper **22** to be discarded or recycled. In some embodiments, in order to reduce maintenance due to potential blockage in the underflow lines **34d**, **34e**, the system **200** may exclude the underflow lines **34d**, **34e** with the underflow from separators emptying directly into an underflow hopper **22**. The overflow from the second separator **18e** comprises twice cleaned liquid that may be directed via overflow line **32e** to the outlet **24**. Although two separators **18d**, **18e** are shown in FIG. 2, the system **200** may include as many separators as desired, wherein each separator includes its own supply tank to collect liquid from the previous separator and a pump to direct the liquid from the tank.

[0030] In one or more embodiments, each of the separators **18d**, **18e** in the system **200** may be the

same or different. For example, the first separator **18d** may be larger than the second separator **18e** such that the first separator **18d** is configured to remove larger insolubles and the second separator **18e** is configured to remove smaller remaining insolubles.

[0031] In some embodiments, either or both of the separators **18d**, **18e** of the system **200** may comprise a plurality of separators connected in parallel (as in FIG. 1). In any of the foregoing embodiments, the separators **18d**, **18e** may be configured to operate at the same rate in order to maintain constant liquid levels throughout the system **200**. For example, the first separator **18d** may comprise a single separator configured to operate at a first rate and the second separator **18e** may comprise to separators each configured to operate at half of the first rate.

EXAMPLES

Example 1—Turbidity Analysis

[0032] Three water samples were taken from a scalders every hour over four production days—two day using the separator system described here and two control days without the separator system. Each sample was analyzed for turbidity for a total of 99 samples (**54** example samples and 45 control samples). As summarized in Tables 1 and 2 below, turbidity differed across trial days in several dimensions: mean daily turbidity was lower in treatment days compared to control days, as was maximum daily turbidity and the daily variance in turbidity.

TABLE-US-00001 TABLE 1 Daily turbidity measures (in Nephelometric Turbidity Units (NTU))

	Mean	Min	Max	Std. Dev.
Example 1 Day 1	109.1	23.4	266.7	47.4
Example 1 Day 2	190.0	24.6	295.7	82.9
Control Day 1	242.7	6.4	418.0	110.0
Control Day 2	248.6	43.1	393.3	116.6

TABLE-US-00002 TABLE 2 Difference in turbidity measures by treatment (in NTU)

	Control	Example 1	Difference	p-value	Mean	Std. Dev.
Example 1 Difference			245.9	145.1	-100.6	$p < 0.01$ (-41.0%)
Example 1 Difference			112.5	76.6	-35.9	$p < 0.01$ (-31.9%)

[0033] In Table 2, the difference is statistically significant at $p < 0.05$.

[0034] During the above tests, it was observed that water turbidity increased very quickly within the first two hours on control days from <100 NTU to 225 NTU and continued to increase throughout the rest of the day, reaching a high of ~340 NTU by the end of the day. Water turbidity increased much more slowly on days using the separator system, reaching only 125 NTU after the first two hours; it continued to rise slowly, reaching a maximum of 200 NTU (40% lower than the maximum on control days) between 12:00 noon and 1:00 pm, and fell back to 100 NTU by the end of the day. Water turbidity had statistically significantly less variation on days using the separator system (i.e., measurements were more tightly grouped together). The results of Example 1 are depicted in FIG. 3, wherein the top line represents the control data, and the bottom line represents the data using the separator system.

Example 2—Microbial Analysis

[0035] The samples described in Example 1 were tested for bacterial loads, namely Aerobic Plate Count (APC) and Enterobacteriaceae count (EB). The results are summarized in Table 3 below. The results are also shown in FIG. 4 and FIG. 5. As compared with the control days, the days using the separator system provided statistically significant reductions in both APC and EB loads in the scalders water.

TABLE-US-00003 TABLE 3 Scalders water bacterial results

	Difference between Control and Example 2	Std. Error	p-value
Log(APC)	-0.65	logs 0.15	$p < 0.0001$
Log(EB)	-0.44	logs 0.22	$p = 0.0452$

Example 3—Solids Analysis

[0036] The samples described in Example 1 were analyzed for solid content. The percent solids differed significantly and was lower in Example 3 as compared to the control group. The results are summarized in Table 4 below and shown in FIG. 6.

TABLE-US-00004 TABLE 4 Difference in water solids measurements by treatment

	Control	Example 3	Difference	p-value
% Solids	0.106	0.083	-0.023	$p = 0.0174$ (-21.7%)

Example 4

[0037] Three carcass samples were taken at hot rehang every hour over four production days—two day using the separator system described here and two control days without the separator system. Each sample was analyzed for APC, EB, *Campylobacter*, and *Salmonella* for a total of 99 samples each (54 separator system samples and 48 control samples).

[0038] After aggregating bacterial results across days and comparing the separator system performance to the control performance there was a numerical reduction in all measures. The effect of the separator system is shown in Table 5 below as the estimated reduction in bacterial loads (or prevalence, in the case of *Salmonella*) due to the separator system when incoming variables are held constant. Statistically significant reductions in both EB and *Campylobacter* loads at hot rehang were found for the separator system.

TABLE-US-00005 TABLE 5 Hot rehang bacterial results Effect of Separator System Std. Error p-value Log(APC) -0.11 logs 0.14 p = 0.4560 Log(EB) -0.35 logs 0.20 p = 0.0443 Log(Campy) -0.49 logs 0.13 p = 0.0006 Salmonella -3.2% 4.7% p = 0.5001

[0039] Turbidity results from Example 1 suggest that the differences in water quality due to the separator system become statistically significant after approximately 90 minutes of use. Without being bound by theory, it is hypothesized that prior to 90 minutes the water quality in the scalding is still “clean” due to daily water change and limited use. After 90 minutes, the control scalding becomes substantially more turbid compared to the separator system-enabled scalding and continues to decline in quality over the course of the day. Thus, the “true” effect of the separator system becomes identifiable within the second hour of testing.

[0040] Based on the foregoing, the hot rehang results from Table 5 were restricted to carcasses processed during the period of time (most of the day) in which the separator system-enabled scalding was significantly less turbid than the control and the above analysis was repeated, as shown in Table 6 below. It was found that a greater reduction in load for both outcomes that were statistically significant in the prior analysis (EB and *Campylobacter*) and the results were again statistically significant.

TABLE-US-00006 TABLE 6 Hot rehang bacterial results after separator system improves turbidity Effect of Separator System Std. Error p-value Log(APC) -0.23 logs 0.18 p = 0.2100 Log(EB) -0.58 logs 0.28 p = 0.0465 Log(Campy) -0.53 logs 0.17 p = 0.0030 Salmonella -1.6% 6.7% p = 0.8112

[0041] Interestingly, the clearest separation in bacterial metrics occurs between 10:00 and 14:00 for APC, EB, and *Campylobacter*, as shown in FIG. 7, FIG. 8, FIG. 9, and FIG. 10.

[0042] A system for processing food, such as meat, poultry, or vegetables, has been disclosed herein. The system includes a process that uses water and produces wastewater comprising insoluble solids; a pump configured to receive and pressurize the wastewater from the process; and a plurality of centrifugal separators connected in parallel and configured to receive the wastewater from the pump. The separators are each configured to separate out at least a portion of the insoluble solids from the wastewater to produce a cleaned wastewater.

[0043] The system may include any combination of the following features: the insoluble solids comprise poultry or other feed, dirt, litter, fecal matter, viscera, entrails, skin, fats, and/or digesta; one or more sensors in communication with a control panel, the control panel comprising a display configured to display data from the one or more sensors; wherein the one or more sensors comprise a temperature sensor, a pressure sensor, a flow rate sensor, or a combination thereof; a sample port positioned between the pump and the separators, the sample port configured to receive and analyze a sample of the wastewater; wherein the sample port is configured to measure turbidity, pH, solids content, microbial content, or a combination thereof and to transmit measurements to the control panel; wherein the control panel comprises a variable frequency drive configured to control a speed of the pump and wherein the variable frequency drive is controllable via the control panel; wherein the process is an inside outside bird washer or other food washing process, a chiller, an antimicrobial application process, or a scalding; wherein the cleaned wastewater is recycled into the

process; an additive supply configured to introduce an antimicrobial, a pH adjuster, and/or fresh water into the cleaned wastewater.

[0044] An agricultural processing system has been described herein. The system includes a process that uses water and produces wastewater comprising insolubles; a first pump configured to receive and pressurize the wastewater from the process; a first centrifugal separator configured to receive the wastewater from the first pump and separate out at least a portion of the insolubles from the wastewater to produce a once-cleaned wastewater; a tank configured to receive and hold the once-cleaned wastewater; a second pump configured to receive and pressurize the once-cleaned wastewater from the tank; and a second centrifugal separator configured to receive the once-cleaned wastewater from the second pump and separate out at least a portion of the insolubles from the once-cleaned wastewater to produce a twice-cleaned wastewater.

[0045] The system may include any combination of the following features: wherein the first centrifugal separator is configured to remove insolubles having at least a first diameter and the second centrifugal separator is configured to remove solid particles having at least a second diameter; and wherein the second diameter is smaller than the first diameter; wherein at least one of the first centrifugal separator or the second centrifugal separator comprises a plurality of centrifugal separators connected in parallel.

[0046] A method of processing food has been described herein. The method includes contacting food workpieces with water to produce wastewater comprising insoluble solids dislodged from the workpieces; using a pump to direct the wastewater to a plurality of centrifugal separators; using the separators to remove at least a portion the insoluble solids from the water to produce a cleaned wastewater; and recycling the cleaned wastewater.

[0047] The method may include any combination of the following features: wherein the insoluble solids comprise poultry or other feed, dirt, litter, fecal matter, viscera, entrails, skin, fats, and/or digesta; wherein contacting the food workpieces with water comprises immersing the workpieces in a chiller tank; wherein contacting the food workpieces with water comprises immersing the workpieces in a scalding tank; wherein the plurality of centrifugal separators are connected in series and each of the separators is identical; wherein recycling the cleaned wastewater comprises returning the cleaned wastewater to the chiller tank; wherein recycling the cleaned wastewater comprises returning the cleaned wastewater to the scalding tank.

[0048] The above specific example embodiments are not intended to limit the scope of the claims. The example embodiments may be modified by including, excluding, or combining one or more features or functions described in the disclosure. The description of the present disclosure has been presented for purposes of illustration and description but is not intended to be exhaustive or limited to the embodiments in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosure. The illustrative embodiments described herein are provided to explain the principles of the disclosure and the practical application thereof, and to enable others of ordinary skill in the art to understand that the disclosed embodiments may be modified as desired for a particular implementation or use. The scope of the claims is intended to broadly cover the disclosed embodiments and any such modification.

Claims

1. An agricultural processing system, comprising: a process that uses water and produces wastewater comprising insoluble solids; a first pump configured to receive and pressurize the wastewater from the process; a first centrifugal separator configured to receive the wastewater from the first pump and separate out at least a portion of the insoluble solids from the wastewater to produce a once-cleaned wastewater; a tank configured to receive and hold the once-cleaned wastewater; a second pump configured to receive and pressurize the once-cleaned wastewater from

the tank; and a second centrifugal separator configured to receive the once-cleaned wastewater from the second pump and separate out at least a portion of the insoluble solids from the once-cleaned wastewater to produce a twice-cleaned wastewater.

2. The system of claim 1, wherein the first centrifugal separator is configured to remove insoluble solids having at least a first diameter and the second centrifugal separator is configured to remove insoluble solids having at least a second diameter; and wherein the second diameter is smaller than the first diameter.

3. The system of claim 1, wherein at least one of the first centrifugal separator or the second centrifugal separator comprises a plurality of centrifugal separators connected in parallel.

4. A method of processing food, comprising: contacting food workpieces with water to produce wastewater comprising insoluble solids dislodged from the workpieces; using a pump to direct the wastewater to a plurality of centrifugal separators; using the separators to remove at least a portion of the insoluble solids from the water to produce a cleaned wastewater; and recycling the cleaned wastewater.

5. The method of claim 4, wherein the insoluble solids comprise animal feed, dirt, litter, fecal matter, viscera, entrails, skin, fats, and/or digesta.

6. The method of claim 5, wherein contacting the workpieces with water comprises immersing the workpieces in a chiller tank.

7. The method of claim 6, wherein recycling the cleaned wastewater comprises returning the cleaned wastewater to the chiller tank.

8. The method of claim 5, wherein contacting the workpieces with water comprises immersing the workpieces in a scalding tank.

9. The method of claim 8, wherein recycling the cleaned wastewater comprises returning the cleaned wastewater to the scalding tank.

10. The method of claim 4, wherein the plurality of centrifugal separators is connected in series and each of the separators is identical.
